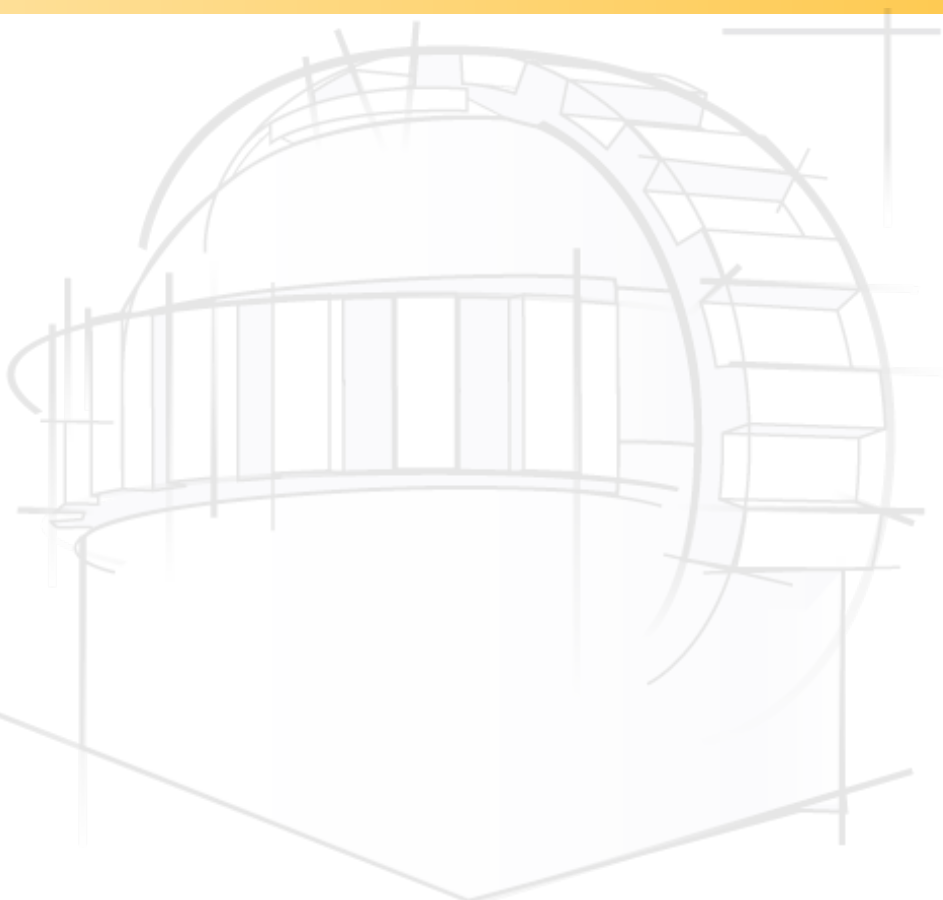
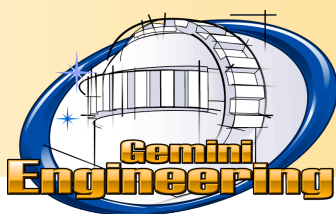


09B TCC User Interface Changes

GE-XX-YY-ZZ-XXX-XX



Issued By: SW Engineering Group



Created By: Javier Lührs

Approved By: [Click here to enter text.](#)

Group Manager: Vasudeva Upadhya

Version History

REVISION CHART			
Version	Author(s)	Description of Version	Date Completed
1.0	Javier Lührs	First version.	June 16, 2009

Table of Contents

VERSION HISTORY	3
TABLE OF CONTENTS	4
INDEX OF TABLES	5
INDEX OF FIGURES	6
1 PURPOSE	7
2 APPLICABLE DOCUMENTS	8
3 LIST OF ACRONYMS	9
4 INTRODUCTION	10
4.1 GeMS Guide Stars	10
5 TCC RELEASE 2009B	11
5.1 Field Configuration Panel	11
5.1.1 Selecting a Target for a Guider or the Telescope	12
5.1.2 Modifying Target List Entries	13
5.1.3 Adding Target List Entries	13
5.1.4 Removing Target List Entries	13
5.1.5 Modifying Targets	13
5.1.6 Creating New Targets	14
5.1.7 Swap Science/Guide Target	14
5.1.8 Guide Configuration	15
5.1.9 Other Field Parameters and Actions	15
5.2 Other Changes	15
6 KNOWN ISSUES	16

Index of Tables

Table 1:Target tag list.

12

Index of Figures

Figure 1: New Field Configuration Panel.	11
Figure 2: Input fields for Right Ascension - Declination coordinates.	13
Figure 3: Input fields for Azimuth - Elevation coordinates.	14
Figure 4: Input fields for Non-sidereal objects.	14

1 Purpose

This document describes the changes introduced in the TCC user interface for the 2009B release.

2 Applicable Documents

N.A.

3 List of Acronyms

This is a list of acronyms used through this document.

GeMS	Gemini MCAO System
MCAO	Multi-Conjugate Adaptive Optics
OIWFS	On-Instrument Wave Front Sensor
PWFS	Peripheral Wave Front Sensor
TCC	Telescope Control Console
TCS	Telescope Control System
OT	Observing Tool
AO	Adaptive Optics
GSAOI	Gemini South Adaptive Optics Imager
ODGW	On-Detector Guide Window

4 Introduction

The 2009B Release of TCC is the first version to introduce support for the Gemini MCAO System. This new system brings in several new guide stars that must be configured through TCC, in addition to the current PWFS (1 and 2), OIWFS and Altair. The style of the user interface used until now in TCC for configuring the different targets could not deal properly with the increase of configurable targets. Thus, a redesign of the user interface was necessary.

4.1 GeMS Guide Stars

GeMS introduces seven new guide stars, of two different kinds. They are:

- Canopus 1, 2 and 3: Used by GeMS for Tip-Tilt and Slow Focus correction.
- ODGW 1, 2, 3 and 4: Available only with GSAOI, these are used for Tip-Tilt and flexure correction.

Up to 4 stars can be used at the same time.

5 TCC Release 2009B

5.1 Field Configuration Panel

Almost all the user interface changes affect the Field Configuration Panel. The new panel is shown in Figure 1.

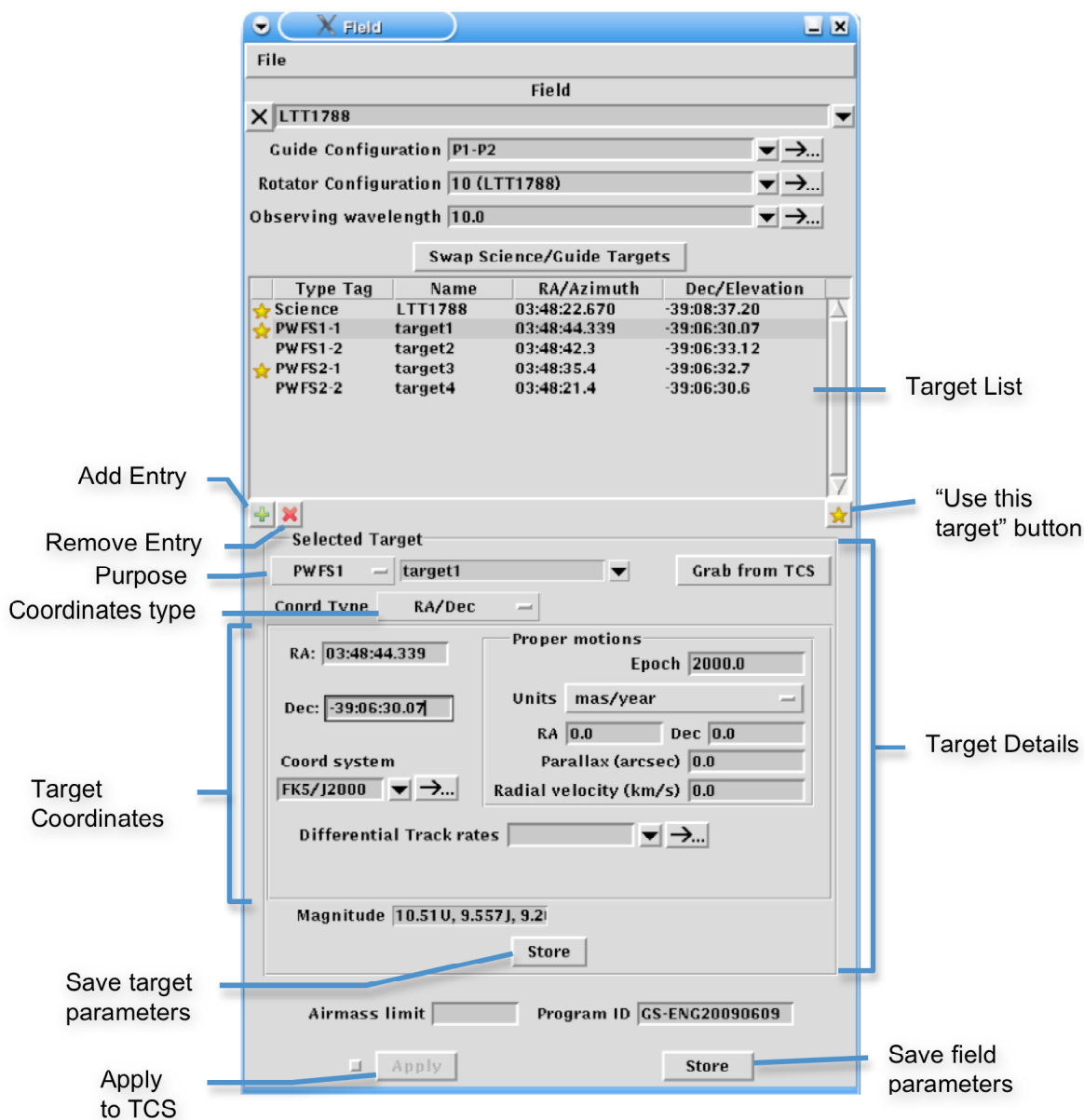


Figure 1: New Field Configuration Panel.

The tabs for editing the target for the telescope and the guiders were removed. Instead, the Field Configuration Panel presents a list of targets and its intended use, much like the one used in the OT. This Target List shows in each line a target and its use (guide star or used to point the telescope). The second column denotes such purpose with a tag. The tags and their meaning are listed in Table 1. The third column identifies the target, and the rest of the columns show its coordinates.

Table 1: Target tag list.

Tag	Use
Science	Base telescope target.
User	Base telescope target (optional).
PWFS1	Peripheral wave front sensor 1 target.
PWFS2	Peripheral wave front sensor 2 target.
OIWFS	On-instrument wave front sensor target.
AOWFS	Target for AO system.
CWFS1	Target for CANOPUS wave front sensor 1.
CWFS2	Target for CANOPUS wave front sensor 2.
CWFS3	Target for CANOPUS wave front sensor 3.
ODGW1	Target for on-detector guide window 1.
ODGW2	Target for on-detector guide window 2.
ODGW3	Target for on-detector guide window 3.
ODGW4	Target for on-detector guide window 4.

It is possible to have more than one alternative target for pointing the telescope, or for configuring each one of the guiders. A unique number is added to the tag to differentiate the list entries in that case. For example, if there are two possible targets to use for PWFS2, they will appear in the target list as PWFS2-1 and PWFS2-2.

One target must be marked from between all the alternatives for a particular guider as the one to actually be used for that guider. This target is denoted in the target list by a yellow star next to the target's tag.

The main target for the observation is identified as "Science". Other alternatives for pointing the telescope are identified with the tag "User". A star next to the target's tag in the target list denotes which one will be used when configuring the telescope, just as the guiding targets.

As mentioned before, each row of the Target List presents a target and its possible use. As a target can have more than one intended use, it can appear in more than one row. In that case the rows will differ only in the tag. Any change to the target parameters will be reflected in all the rows where it appears.

Targets not assigned to any guider nor the telescope will not appear in the target list. They can still be assigned to a guider selecting them in the drop-down list in the Target Details area, below the Target List.

Most operations require selecting an entry in the Target List. The selected row is identifiable by a darker background.

5.1.1 Selecting a Target for a Guider or the Telescope

To choose a target for one of the guiders (or the telescope), first select the target in the target list, and then press the button marked with a yellow star (identified in Figure 1 as the "Use this target" button). That will mark the target as the one to be used by the guider corresponding to its tag. If the selected

target already was marked, it will be unmarked, and no target will be configured for that particular guider. The only exception to this behavior is the telescope target. Any field configuration must define at least the target to be used for the telescope. The software takes some actions to enforce this. If you try to unmark a “User” target, it will be unmarked, but the “Science” target will be marked instead. It is not possible to unmark the Science target, except by marking a “User” target.

5.1.2 Modifying Target List Entries

You can edit any entry in the Target List and change its target or its tag. First, select the row from the Target List. Then in “Selected Target” area, use the “Purpose” selector to set a new guider. For use the target to point the telescope, use the types Science or User. To change another target, just select it from the drop-down list, or write its name. If a target with that name does not exist, a new target will be created.

To confirm the changes, press the “Store” button in the target area.

There are some particular behaviors for the Science entry. There must be one and only one such entry. You cannot change the tag for this one. If you select the Science tag for another entry, it will replace the previous entry.

5.1.3 Adding Target List Entries

To add a new entry in the Target List, just press the “Add Entry” button (identified by a green cross). If the Target List is empty, the first entry will always be for the “Science” target. Otherwise, the new entry will be created with the same tag as the previously selected.

A new empty target is created for each new entry.

5.1.4 Removing Target List Entries

To remove an entry in the target list, first select the entry, and then press the “Remove Entry” button (identified by a red X).

The Science target entry cannot be removed.

5.1.5 Modifying Targets

To modify a target, just select any entry on the Target List that has that target. The parameters of the target will appear in the “Selected Target” area, where they can be edited. The “Coordinates Type” states how the coordinates for the target will be defined. The area below this control changes to present the target parameters according to the coordinates type selected, as shown in Figure 2, Figure 3 and Figure 4.

The screenshot shows a software interface for configuring a target. It includes input fields for Right Ascension (RA) and Declination (Dec) in J2000 coordinates, a dropdown for the coordinate system (set to FK5/J2000), and a section for proper motions with fields for Epoch, Units (mas/year), RA, Dec, Parallax (arcsec), and Radial velocity (km/s). There is also a field for Differential Track rates.

RA: 03:48:44.339	Proper motions	
Dec: -39:06:30.07	Epoch 2000.0	
Coord system	Units mas/year	
FK5/J2000 ▼ →...	RA 0.0 Dec 0.0	
	Parallax (arcsec) 0.0	
	Radial velocity (km/s) 0.0	
Differential Track rates	▼ →...	

Figure 2: Input fields for Right Ascension - Declination coordinates.

Figure 3: Input fields for Azimuth - Elevation coordinates.

Figure 4: Input fields for Non-sidereal objects.

It is not necessary to input orbit parameters for major planets or the Moon. They can be just selected from the drop-down list for non-sidereal objects.

The coordinates of the target can be retrieved from the current TCS configuration. To do that, just press the button marked as “Grab from the TCS”. The coordinates will be retrieved from the currently set parameters for the telescope or one of the guiders. The tag on the selected entry in the Target List determines the actual source of the values. This operation only works for targets defined in RA/Dec or Az/El coordinates.

To finalize the changes, you must press the “Store” button in the target area. If you fail to do that, and select another target, your changes will be lost.

Targets are identified by its name. The consequences of this are that if you write a new name in the text box of the pull-down list and press the “Store” button, you will create a new target. If the name coincides with an already existing target, it will be overwritten.

5.1.6 Creating New Targets

As mentioned before, a new target is created each time a new entry is added to the Target List, or when you change the name of a target to one that does not previously existed.

5.1.7 Swap Science/Guide Target

This operation works as before, except that now it is possible to select any available guide target for this operation. The first step is to select a guide target from the Targets List. Only market targets (indicated by the star icon) are valid. After selecting the desired star, just press the “Swap Guide/Science” button. The guide and the telescope target will be marked with a “swapped” icon until they are restored.

The drawback of this new feature is that now the operation requires one extra step.

5.1.8 Guide Configuration

The Guide Configuration selector includes two new configurations, introduced to support GeMS. They are:

- GeMS: Activates guiding for all the configured GeMS guide stars.
- GeMS-OI: Activates guiding for CANOPUS guide stars and OIWFS.

5.1.9 Other Field Parameters and Actions

All other field parameters remain the same. After changing any field parameter it is necessary to press the “Store” button to confirm the changes (The one next to the “Apply” button, don’t confuse with the “Store” button in the target area). Otherwise, they will not be acknowledged.

To configure the TCS with the parameters of the current selected field, press the “Apply” button. The “Apply” button will store unsaved changes to the field configuration before applying it to TCS.

5.2 Other Changes

There are some other changes to the TCC besides the Field Configuration panel. They are listed below:

- There are two new instruments available in the instrument selector: GSAOI and FLAMINGOS2.
- Likewise, there are two new instrument origins: gsaoi and flamingos2.
- The OIWFS filter selector in the Acquisition and Guiding panel has a new filter for FLAMINGOS-2 OIWFS.
- The format of the XML files containing TCS configurations was modified to support the list of alternative targets for each guider. TCC can still read the old format.

6 Known Issues

This is a list of know issues of the new TCC.

1. The new GeMS guide stars are missing from the Guide Configuration panel.
2. They are not shown in the Preview, neither.
3. TCS support the seven new guide stars through 4 “virtual” guide stars, called G1, G2, G3 and G4. The TCS error messages use the name of the virtual guide star instead of the GeMS guide star it configures, so it is not always easy to know which guide star an error message refers to. As a rule, TCC configures guide stars G1 to G4 with the available GeMS stars in this order: CWFS1, CWFS2, CWFS3, ODWG1, ODWG2, ODWG3, ODWG4.
4. In the previous version of TCC, a Track Rate panel could be invoked from any of the guide target tabs. The Track Rate panel had an “Apply” button used to apply this track rate on the corresponding guide target configuration in TCS. That was possible because the target edited was always the one to be applied for the selected guider. But now there can be multiple alternatives for a given guider, and any of those targets can be edited, not only the currently selected for that guider. This means the “Apply” button in the Track Rate panel not always make sense. For this reason it was decided to remove that button. The same situation affects the Coordinate System panel. Track rates and coordinate systems are still applied when applying the field configuration.
5. Target names cannot contain characters that can be misinterpreted by Tcl, like dollar signs (\$), or square brackets ([]).
6. The GUI does not prevent creating duplicate entries in the Target List.